The Impact of Emergency Department Shift Scheduling on Clinician Fatigue and Well-being: A Qualitative Study

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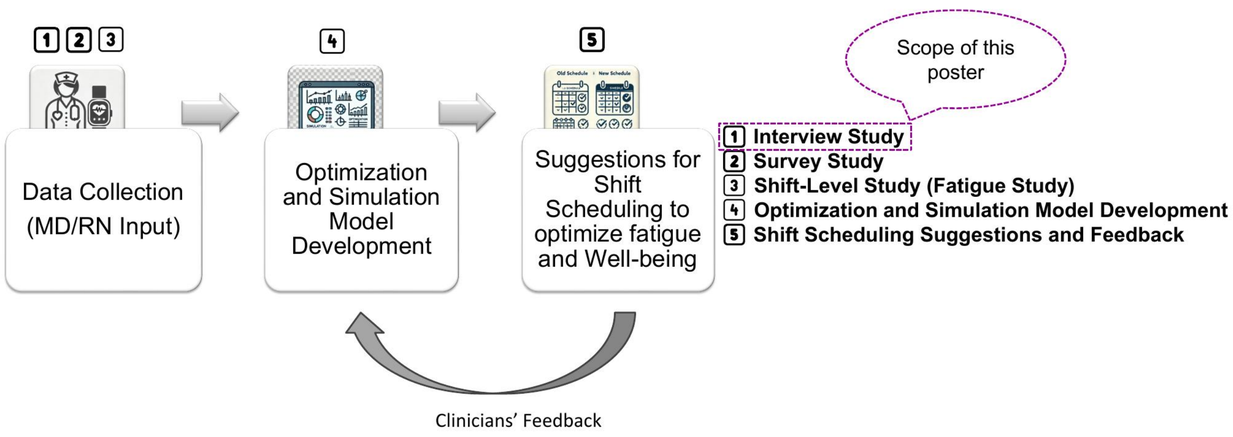


Figure 1. Phases of the research project

Purpose Statement

- ✓ Emergency Department (ED) clinicians face heightened risks of **fatigue, burnout,** and **reduced well-being**.
- ✓ These challenges are closely linked to the **structure and distribution** of shift schedules.
- ✓ **Inflexible scheduling**, **limited recovery time**, and **circadian misalignment** (e.g., night or rotating shifts) disrupt rest and recovery cycles.
- ✓ Resulting impacts include **diminished performance**, **increased error risk**, and **higher workforce attrition**.
- ✓ This **qualitative interview study** explores how **shift length, timing, distribution,** and **schedule flexibility** influence clinician **fatigue, burnout, well-being,** and **scheduling preferences**.

Methods

❑ Participants:

- 13 physicians (5 leaders) and 18 nurses (2 leaders) from three Prisma Health EDs participated in a 30-minute-long one-to-one semi-structured interview.

❑ Recruitment:

- Conducted through departmental announcements and direct invitations
- Study protocol approved by the **Prisma Health Institutional Review Board**

❑ Interview Design:

Semi-structured interviews guided by a standardized protocol. Topics included:

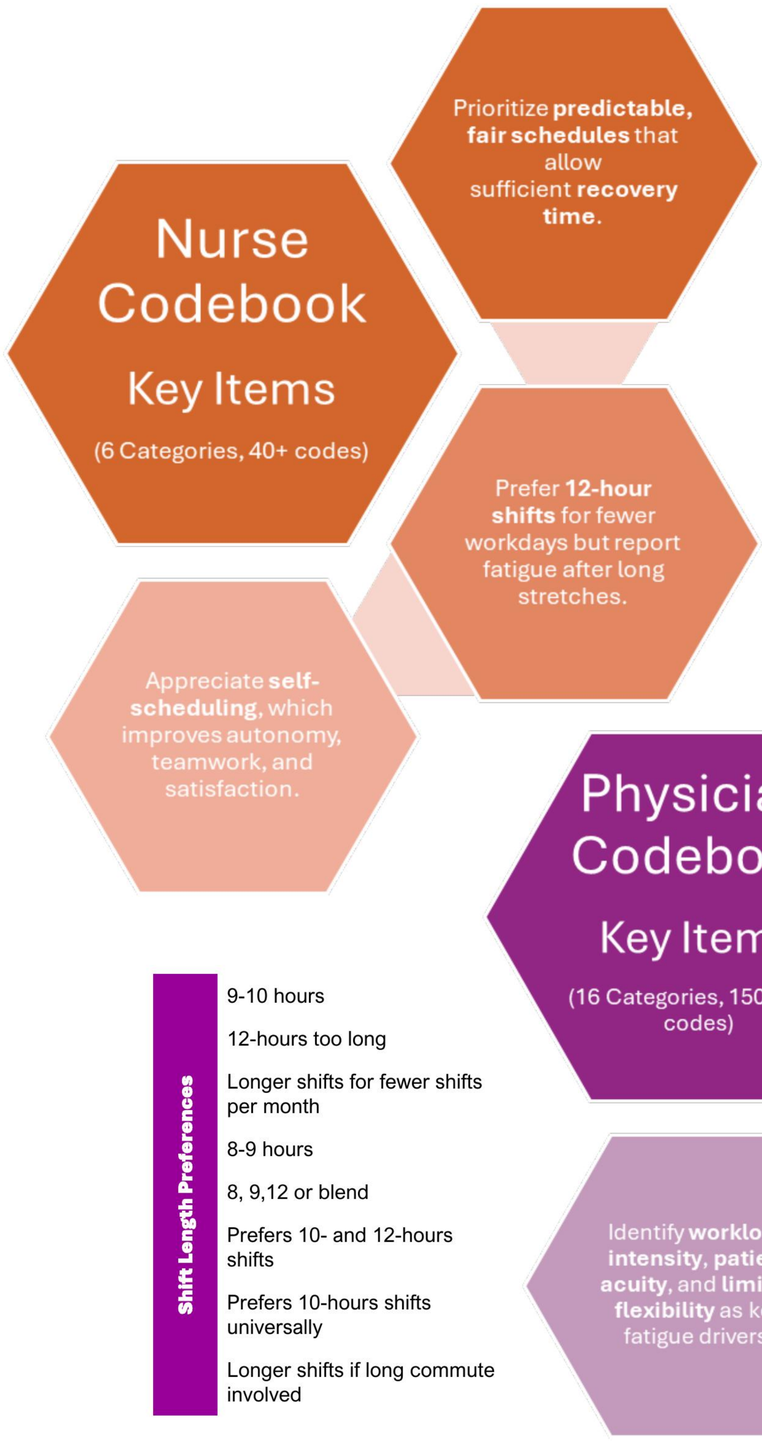
- Ideal schedule characteristics
- Effects of current schedules on fatigue and recovery
- Preferences for nights and weekends
- Flexibility in requesting or swapping shifts
- Perceptions of a "biologically fatigue-optimized" schedule

❑ Data Processing & Analysis:

- Transcripts were cleaned, verified against audio, and coded using a collaboratively developed codebook.
- Coding teams included **three members per group** (physician and nurse).
- **Interrater reliability** is being established through dual coding of a subset, followed by consensus achieved through iterative discussions.
- One will code all nurse transcripts; another will code all physician transcripts.

Results

Codebook Development



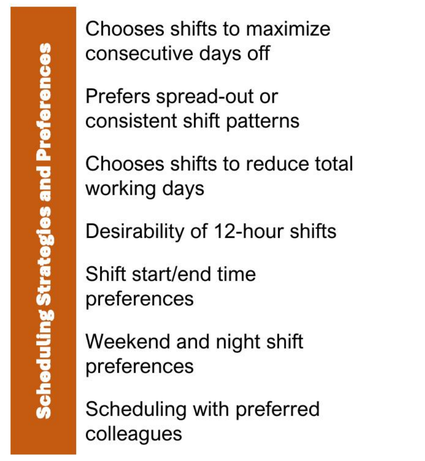


Figure 2. A sample code category and corresponding codes from the nurse codebook (right), and a summary of the nurse codebook (left).

Figure 3. A sample code category and corresponding codes from the physician codebook (left), and a summary of the physician codebook (right).

Shared Drivers

- ✓ Both groups link **fatigue** to **unpredictable shifts**, **high workload**, and **insufficient recovery**.
- ✓ Agree that **control**, **consistency**, and **recovery opportunities** are essential for sustaining **well-being** and **performance**.

Conclusion

- ❖ Coding and thematic analysis are ongoing to refine emerging findings.
- ❖ Clinician fatigue and well-being are strongly influenced by how schedules are structured and managed.
- ❖ Nurses and physicians share common fatigue factors but differ in flexibility, preferences, and recovery needs.
- ❖ Finding of this interview will help to create better survey questions for the next stage of the project.
- ❖ These insights will support developing fair, flexible, and fatigue-aware scheduling and rostering models to maximize the well-being of the clinicians.

For references related to this poster and for any questions, please contact me at mdzahii@clemson.edu